

# VIK SHIRVAIKAR

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Artificial intelligence engineer and machine learning Ph.D. with experience translating complex models into production systems. Technical expertise in forecasting, optimization, Bayesian statistics, causal inference, and uncertainty quantification, with applications spanning finance, healthcare, energy, and sports.

## EDUCATION

**University of Oxford**, September 2025

Ph.D. (Doctor of Philosophy) in Statistics and Machine Learning

*Dissertation title: Prediction-powered machine learning for model selection and uncertainty*

- Created a novel framework for Bayesian uncertainty quantification via iterative prediction
- Developed an R/C++ package for causal subgroup analysis of clinical trials (working with Novo Nordisk)
- Authored research papers across a range of other fields, including deep learning, causal reasoning in large language models (LLMs), reinforcement learning, and public policy evaluation

**The University of Texas at Austin**, May 2019

B.S.A. (Bachelor of Science and Arts) in Mathematics, B.A. (Bachelor of Arts) in Economics, 4.0/4.0 GPA

## WORK EXPERIENCE

**Gyre Energy** – Founding AI/ML Engineer, September 2025 to Present – *London, UK*

*Early-stage startup automating control systems to optimize energy efficiency for commercial buildings*

- Built core AI products end-to-end, from model design to backend architecture and cloud implementation, with live deployments across three continents delivering benchmarked energy savings of up to 40%
- Engineered simulation, forecasting, and optimization tools for automated real-world decision-making
- Developed custom Bayesian approaches for uncertainty quantification from noisy operational data

**Various Clients** – Independent Sports AI/ML Consultant, December 2024 to June 2025 – *Oxford, UK*

- Developed a proprietary athlete valuation algorithm to support data-driven contract negotiations, bringing together on-field performance data and off-field digital presence metrics
- Designed and implemented a geospatial forecasting engine for new football stadium development, integrating demographic, transportation, and economic data to project site-specific revenue streams

**Integra FEC** – Senior Forensic Data Analyst, March 2019 to July 2021 – *Austin, Texas, USA*

*Leading data-driven litigation consulting firm for financial fraud prosecution and enforcement*

- Conducted fraud investigations for U.S. government clients, specializing in consumer protection, structured products, and commercial banking, with largest case settling for over \$1.5 billion
- Designed statistical sampling frameworks and data visualizations for use in legal proceedings
- Built a Google Cloud Platform pipeline from scratch for live currency tracking and auditing

## STRENGTHS AND SPECIALITIES

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|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages and libraries  | Python, R, Julia, C++, Java, Stan, TensorFlow, PyTorch, SQL, Stata, SAS                                                                            |
| Tools and infrastructure | Google Cloud Platform, Microsoft Azure, Git, PostgreSQL, Docker, FastAPI                                                                           |
| Statistical methods      | Bayesian modeling, causal inference, probabilistic programming, econometrics, time-series analysis, model selection, experimental design           |
| AI and ML systems        | Reinforcement learning, large language models (LLMs), AI agent workflows, deep learning, digital twins, optimization and control, model deployment |